Laplace Transform Table

Forward and inverse pairs: read each row in either direction.

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} e^{-st} f(t)\, dt$$

Common transform pairs

$u(t)$: unit step; $\delta(t)$: unit impulse.

No.	Time domain $f(t)$	s domain $F(s)$
1	$\delta(t)$	1
2	$u(t)$	$\dfrac{1}{s}$
3	$t\,u(t)$	$\dfrac{1}{s^2}$
4	$t^n u(t) \quad (n = 0, 1, 2, \ldots)$	$\dfrac{n!}{s^{n+1}}$
5	$e^{-at} u(t)$	$\dfrac{1}{s+a}$
6	$te^{-at} u(t)$	$\dfrac{1}{(s+a)^2}$
7	$t^n e^{-at} u(t)$	$\dfrac{n!}{(s+a)^{n+1}}$
8	$\sin(\omega t) u(t)$	$\dfrac{\omega}{s^2+\omega^2}$
9	$\cos(\omega t) u(t)$	$\dfrac{s}{s^2+\omega^2}$
10	$e^{-at}\sin(\omega t) u(t)$	$\dfrac{\omega}{(s+a)^2+\omega^2}$
11	$e^{-at}\cos(\omega t) u(t)$	$\dfrac{s+a}{(s+a)^2+\omega^2}$
12	$\dfrac{1-e^{-at}}{a}\, u(t)$	$\dfrac{1}{s(s+a)}$
13	$\dfrac{e^{-at}-e^{-bt}}{b-a}\, u(t) \quad (a \neq b)$	$\dfrac{1}{(s+a)(s+b)}$
14	$u(t-T)$	$\dfrac{e^{-Ts}}{s}$
15	$u(t) - u(t-T)$	$\dfrac{1-e^{-Ts}}{s}$
16	$1 - e^{-\zeta\omega_n t}\left[\cos(\omega_d t) + \dfrac{\zeta\omega_n}{\omega_d}\sin(\omega_d t)\right]$ $\omega_d = \omega_n\sqrt{1-\zeta^2}, \quad 0 < \zeta < 1$	$\dfrac{\omega_n^2}{s\,(s^2+2\zeta\omega_n s+\omega_n^2)}$

Parameters: n is a nonnegative integer; $a, b > 0$, $\omega, \omega_n > 0$, and $T \geq 0$. All unmarked signals are causal. Row 16 is the unit-step response of a unity-gain underdamped second-order system.

Laplace Transform Properties

Use these operations before looking up an inverse pair.

$$F(s) = \mathcal{L}\{f(t)\}, \quad G(s) = \mathcal{L}\{g(t)\}$$

Property	Time domain	s domain / result
Linearity	$Af(t) + Bg(t)$	$AF(s) + BG(s)$
First derivative	$\dot{f}(t)$	$sF(s) - f(0^-)$
Second derivative	$\ddot{f}(t)$	$s^2F(s) - sf(0^-) - \dot{f}(0^-)$
Integration from zero	$\int_0^t f(\tau)\,d\tau$	$\frac{F(s)}{s}$
Exponential shift	$e^{-at}f(t)$	$F(s+a)$
Time delay	$f(t-T)u(t-T)$	$e^{-Ts}F(s)$
Multiply by time	$tf(t)$	$-\frac{dF(s)}{ds}$
Time scaling, $c > 0$	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
Convolution	$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$	$F(s)G(s)$
Initial value theorem	$f(0^+)$	$\lim_{s\to\infty} sF(s)$
Final value theorem	$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} sF(s)$
Differentiate n times	$f^{(n)}(t)$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} \cdot f^{(k)}(0^-)$

Checks before using the table

For inverse transforms, factor or complete the square first. Match the denominator and numerator to a row, then apply linearity. A factor e^{-Ts} means a time delay, not a change in pole location.

The final value theorem needs a finite final value: every pole of $sF(s)$ must lie strictly in the left half-plane. The initial value formula excludes an impulse at $t = 0$. Initial conditions in the derivative rows use 0^-.